Ludovico Furlanetto

Physicist · Complex Systems

CONTACT

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NotLudovico
notludovico

TECHNICAL STACK

Languages
Python, Rust, C++, JavaScript

Data & ML
Polars, PyTorch, Scikit-learn, SQL

Tools
Linux, Git, PostgreSQL, Excel, PowerPoint

CERTIFICATIONS

Machine Learning Specialization
Andrew Ng / Stanford, 01/2025

LANGUAGES

Italian (native)
English (C1, FCE)
French (basic)

INTERESTS

Chess
Mixed Martial Arts

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Profile

Physicist working between stochastic modelling and production code. Drawn to problems where theory has to survive contact with real data.

Professional Experience

Centre d'Économie de la Sorbonne — Paris, France
Research Intern, Modelling Firm Growth — 02/2026 – 06/2026
Supervisor: Prof. Aguirre Lopez

- Built a relative generalized Lotka–Volterra model of firm growth, reproducing growth statistics measured from Compustat firm-level data.
- Derived and implemented a Dynamic Mean-Field Theory (DMFT) solver validated against simulations; open-sourced the Python package behind the results.

Education

Politecnico di Torino — Turin, Italy
M.Sc. Physics of Complex Systems (International Track) — 10/2024 – 07/2026

- **Double degree with Sorbonne Université, Paris.**
- Final grade: 110/110 cum laude.

SISSA (International School for Advanced Studies) — Trieste, Italy
Graduate Coursework in Physics & Data Science — 10/2024 – 02/2025

- Probability, information theory, data science and random matrix theory.

ICTP (International Centre for Theoretical Physics) — Trieste, Italy
Spring College in the Physics of Complex Systems — 02/2026 – 03/2026

- Statistical physics of machine learning, RNNs in neuroscience, stochastic thermodynamics.

University of Bologna — Bologna, Italy
B.Sc. in Physics — 09/2021 – 07/2024

- Final grade: 110/110. Thesis: "Modello di tipo dimero e applicazione al modello di Ising".

Technical Projects

Backtes.to, Financial Analysis SaaS — 02/2026 – Present
Core Developer, 2-person team · backtes.to

- Scaling an investment platform serving 50,000+ monthly visits, built in JavaScript and Next.js.
- Optimizing backtesting engines and Supabase queries over millions of rows of financial data.

Ruut Functions, Rust numerics library
Maintainer · crates.io/crates/ruut-functions

- High-performance Rust library for numerical integration and symbolic differentiation; 10,000+ downloads, built around memory safety and computational speed.

Dynamical System Network (C++) — 02/2024 – 04/2024
Contributor, 7-person team · City Lab, University of Bologna

- Optimized urban traffic flow with complex-network modelling on high-performance C++ infrastructure.
- Deployed traffic-light optimization algorithms for Via Stalingrado, Bologna.

Teaching

Private Tutoring — Italy
Mathematics and Physics, high school students — 2023 – Present

- Coaching high school students through resit exams after a failed year.